

RoboReason: A Real-Hardware Harness for Code-Grounded LLM/VLM Manipulation

Matteo Nini¹

Abstract—Vision-language-action (VLA) models fold perception, reasoning, and control into one policy, but need paired vision-action data far scarcer than the text and image corpora an LLM or VLM is pretrained on. We argue instead for a decoupled Sense-Plan-Act architecture: an LLM/VLM plans in language, and a deterministic code layer grounds the plan’s geometry in sensor measurement, so pretrained knowledge transfers through code without task-specific action data. We present a real-hardware ROS2 harness built on this principle. Deploying it on a UR5cb manipulator surfaced six recurring geometry-grounding bugs invisible to symbolic simulation by construction; the code-grounding layer closes five, and a 120-trial benchmark shows the sixth, a multi-object placement collision, as the dominant residual failure in both an LLM and a camera-grounded VLM arm. We report this taxonomy as a discovered validation of the harness’s premise, not our central claim.

I. INTRODUCTION

A vision-language-action (VLA) model maps perception and instruction directly to low-level control, in one policy trained end-to-end. This is an elegant architecture, but it inherits imitation learning’s oldest bottleneck: the policy only knows what its demonstration data covers, and paired vision-action robot trajectories are expensive to collect at the scale and diversity a general-purpose manipulator needs. A large language model (LLM) or vision-language model (VLM), by contrast, is pretrained on a far broader and cheaper-to-collect corpus, text, code, and image-text pairs at internet scale, and arrives already knowing a great deal that is relevant to manipulation: object affordances, spatial and arithmetic reasoning, ordering and precondition logic, none of it robot-specific, all of it transferable through code rather than through a demonstration dataset [1]–[4]. This is the architectural bet a decoupled Sense-Plan-Act pipeline makes: let the model plan in language, and let a separate execution layer turn that plan into motion.

The bet has a cost. A decoupled pipeline has to ask the model, somewhere, to commit to numbers, a release height, a grasp width, a target position, and nothing in an LLM or VLM’s pretraining grounds those numbers in the physical world it is now supposed to act in. Simulated validation hides this cost almost entirely: in a scripted physics engine or a symbolic environment, an object’s height, its grasp width, and the distance a gripper must travel to release it safely are never actually *measured*, they are asserted, correctly, by construction, so a model’s arithmetic is never tested against a sensor that could contradict it [5]–[7]. On real hardware

none of these quantities are given. They have to be computed, from noisy depth, from a calibrated but imperfect camera, or from a hand-authored scene file that may or may not match the physical table it describes.

We present a real-hardware ROS2 harness built around this architectural choice (§III): a modular Sense-Plan-Act pipeline supporting six reasoning strategies over two grounding modalities (a static-scene LLM planner and a camera-grounded VLM planner), a small library of parameterized manipulation skills, and a deterministic code-grounding layer that overrides model-computed geometry with a sensor- or action-derived value wherever the two could disagree. A companion framework from the same group, RoboReason Lab [8], implements the same class of reasoning strategies over the same kind of pipeline, but entirely inside CopeliaSim; ours, as far as we are aware, is the first version of it run on a real manipulator.

Building and operating this harness on a real UR5cb manipulator is what surfaced the cost described above concretely. Over six development sessions we found a recurring and costly pattern: whenever a planning prompt asked the model to compute a piece of manipulation geometry itself, that computation was often wrong, in ways that stayed silent until the arm physically dropped, crushed, or missed an object. We diagnose six such bugs from production hardware logs (§IV), and close five of them with the harness’s code-grounding layer, which computes the same quantities deterministically from paired pick/release actions and measured depth instead of trusting the model’s own arithmetic (§V). The sixth, a multi-object placement collision, we diagnose but have not yet closed. It turns out to be the dominant one: a 120-trial real-hardware benchmark (§VI) shows it as the residual failure distribution once the other five are fixed, occurring in *both* the LLM and VLM pipelines regardless of which one grounded the scene. We read this taxonomy as a discovered validation of the harness’s central design premise, not as this paper’s main contribution: an LLM/VLM harness needs a code-grounding layer precisely because real deployment keeps asking the model to compute quantities that only a sensor, or an already-committed action, should determine.

The contributions of this paper are threefold: nousep

- A real-hardware ROS2 harness for decoupled LLM/VLM manipulation planning, with a code-grounding layer as an explicit architectural component, presented as the real-hardware counterpart to simulation-only frameworks for the same class of reasoning strategies (§III).

¹The authors are with [Department, University]. E-mail: matteo.nini@[domain].

- A taxonomy of six recurring geometry-grounding bugs, discovered while operating the harness on a real manipulator across six development sessions, five of which the code-grounding layer closes (§IV, §V).
- A 120-trial real-hardware benchmark, methodologically anchored to a directly comparable symbolic-simulation study [5] and using a single unified model for both grounding modalities, showing that the sixth, still-open bug, multi-object placement collision, is the dominant residual failure in *both* arms, and what that implies for evaluating LLM/VLM robot harnesses before hardware deployment (§VI).

II. RELATED WORK

From symbolic task-and-motion planning to LLM-based planning. Robotic planning has long been structured around a Sense-Plan-Act pipeline [9], originally realized through hierarchical and symbolic planners that search a discrete action space for a sequence provably achieving a goal condition [10]. These planners give strong correctness guarantees but require a hand-specified symbolic model of the world and its dynamics. Foundation models have been used to ground natural-language instructions in robotic affordances [11], to act as zero-shot planners [12], and to generate feasible motion plans directly from instructions [13]. A range of reasoning strategies has been proposed on top of these capabilities: reactive, single-step prompting; interleaved reasoning-and-acting (ReAct) [14]; iterative self-critique (Self-Refine) [15], including a long-horizon sequential-planning variant validated at ICRA [16]; search-based deliberation (Tree-of-Thoughts) [17]; and ensemble voting over independently sampled plans (chain-of-thought with self-consistency) [18], [19]. Recent surveys catalogue this space in detail [20]–[24]. Our contribution is orthogonal to this line of work: we do not propose a new reasoning strategy, and hold the reasoning strategy fixed in our evaluation (§VI). Instead, we show that the geometric grounding surrounding *any* of these strategies, how a chosen action’s numeric parameters are computed, is a separate failure axis that persists regardless of which reasoning strategy produced the action.

End-to-end and vision-language-action approaches. An alternative architecture collapses perception, reasoning, and control into a single model, as in vision-language-action (VLA) systems [25]–[28]. Trained or fine-tuned end-to-end, such a policy can in principle learn to compensate for geometric grounding errors on its own. It pays for that with the one resource imitation learning has always been bottlenecked on: paired vision-action demonstration data, collected per embodiment, which stays scarce and narrow next to the text/code/image-text corpora an LLM or VLM is pretrained on. Code-generating policies that couple perception directly to control primitives [1]–[4] sit closer to our own approach: they keep the decoupled Sense-Plan-Act split and let an LLM’s pretrained code and spatial-reasoning knowledge stand in for embodiment-specific demonstrations. A fully end-to-end VLA also gives up the diagnosability a

decoupled pipeline provides: when it fails, it is difficult to determine whether the failure was a misperception, a reasoning error, or a motor-control error [5]. We adopt the decoupled architecture for both reasons at once, data efficiency and diagnosability, since a fully end-to-end system would not permit the diagnosis of §IV at all, only its symptoms.

Simulation and sim-to-real transfer. The sim-to-real gap is a long-studied concern in robot learning, primarily for learned low-level control policies [29]. Physics simulators such as Gazebo [30], MuJoCo [7], and CoppeliaSim [6] narrow this gap for control by modeling contact dynamics and actuation limits [31]. The gap we document is different in kind. It does not concern whether a *learned policy* generalizes from simulated to real dynamics, but whether a *language-model-computed* geometric quantity, one that a symbolic benchmark never forces the model to compute against a contradicting sensor, is trustworthy at all. To our knowledge this failure class has not been previously isolated and named for LLM/VLM robot planners specifically.

Direct comparison. The closest prior work to this paper is Favali et al. [5], which introduces a task/environment complexity taxonomy and TS/TSR/AETS metrics for benchmarking LLM-based robotic planning across six reasoning strategies, evaluated in a symbolic simulator with exact ground truth. We adopt the same evaluation philosophy and, for §VI, the same metric family, but on real hardware, where ground truth for safety and success is not automatically available (§VI details the resulting semi-automatic adaptation). Neither it nor its companion systems paper, RoboReason Lab [8] (§III-B), ever grounds a plan in measured depth or a calibrated camera; both are simulation-only by design, and so structurally unable to produce the failures §IV diagnoses.

III. THE HARNESS

A. Design rationale: code-grounding over end-to-end control

A vision-language-action (VLA) model collapses perception, reasoning, and control into one policy, trained or fine-tuned end-to-end on embodiment-specific demonstration data. That data is the bottleneck: paired vision-action robot trajectories remain far scarcer, and far more expensive to collect, than the text, code, and image-text corpora a general-purpose LLM or VLM is pretrained on. A decoupled Sense-Plan-Act architecture sidesteps this bottleneck in a specific way. If planning is text (or text conditioned on an image), and the resulting plan is a sequence of parameterized skills rather than raw joint torques, then whatever manipulation-relevant knowledge the model already has, spatial reasoning, arithmetic, common-sense physics, ordering and precondition logic, is available zero-shot, through code, without collecting a single task-specific demonstration. This is the same argument prior code-generation-for-robotics work makes [1]–[4]: an LLM that has never seen a robot trajectory has still seen a great deal of code, and a plan expressed as a short structured program is something that knowledge already transfers to.

The cost is that a decoupled architecture asks the model to compute physical quantities, a release height, a grasp width, a target position, that it did not learn from a source with any grounding in the real world at all. §IV is what happens when that computation is trusted uncritically. Our answer, developed over the harness described below, is not to abandon the decoupled architecture but to constrain it: let the model choose *what* to do and *where*, in symbolic terms, and let a deterministic layer compute *how much*, from a sensor measurement or a paired action already in the plan, whenever the plan requires a number the model cannot be trusted to produce on its own.

B. System architecture

The harness is a ROS2 stack organized around the same Sense-Plan-Act pipeline as §II, shown in Figure 1. A task interface (a terminal client or a web GUI) forwards a natural-language command to one of three planner nodes: a static-scene LLM planner, a camera-grounded VLM planner, or a hybrid that runs a VLM scene-grounding call first and hands the resulting scene to the same LLM planner used in static-scene mode. Whichever planner produced it, the plan is validated and sequenced by a plan-manager node, which drives a skill-executor node (a physics-free mock for fast iteration, or the real UR5cb driver) through a small, fixed set of parameterized skills: `approach`, `pick`, `release`, `move_home`, and `wait`. Every planning call supports six interchangeable reasoning strategies, from a single-step reactive baseline to self-consistency voting and tree search, selectable without touching the pipeline itself.

Two design choices are what let this pipeline scale into an actual research tool rather than a one-off demo. First, every runtime parameter, model choice, reasoning strategy, grounding modality, timeouts, robot IP, is a field on one centralized settings object, overridable by an environment variable with no code change, which is what lets §VI’s conditions be defined and reproduced as configuration rather than as branches of the code. Second, every planning call records a per-run debug artifact, the command, the active configuration, the raw model response, the execution log, and, in VLM modes, the camera frame the plan was grounded in, which is what made diagnosing §IV’s failures possible in the first place: each bug there was traced from a hardware failure back to a specific malformed field in a specific logged response, not inferred after the fact. A web GUI supervises the whole stack (including the physical UR driver and the camera service, both of which fail intermittently enough on real hardware to need their own restart logic) as a set of monitored child processes, so that a trial can be launched, watched, and annotated without a terminal per node.

This is, to our knowledge, the first harness of this kind, decoupled LLM/VLM planning with a code-grounding layer, validated on *real* manipulator hardware rather than in simulation. A companion framework from the same group, RoboReason Lab [8], implements the same class of reasoning strategies over the same kind of Sense-Plan-Act pipeline, but entirely inside CoppeliaSim; ours is its real-hardware

counterpart, and, as §IV shows, real hardware surfaces an entire failure class that a simulation-only harness structurally cannot.

IV. GEOMETRY-GROUNDING FAILURES DISCOVERED ON REAL HARDWARE

Operating the harness of §III on real hardware is what turned the design rationale of §III-A from an argument into an observation. This section reports six recurring geometry-grounding bugs, each found on real hardware and each traced to its root cause using the harness’s own per-run debug artifacts (raw model response, config, and, in VLM mode, the captured camera frame). Table I summarizes all six. The remainder of this section develops the two most instructive of the five closed cases in full, since they recurred independently in two different planning pipelines and therefore best illustrate that the underlying pattern is structural, not a one-off implementation slip. It then treats the sixth, still-open bug on its own (§IV-C), since that one turns out to be the dominant failure mode in §VI.

A. Case 1: top-vs-mid-body grasp/release mismatch

Our executor deliberately grasps objects mid-body rather than at their top surface: the pick descent is $d = \min(0.5h, d_{\max})$, where h is the object’s height and $d_{\max}$ is a clamp that prevents the gripper’s rigid mounting body from crashing into tall objects. Centering the grasp mid-body, rather than pinching at the very top edge, produces a more stable hold on typical rigid objects.

The paired *release* action, however, computes how far to lift the object before opening the gripper (the `object_height` parameter, added to `release_position.z` by the executor at execution time) by assuming the object was grasped at its top. This assumption is wrong by construction whenever the pick descended mid-body, which is always. The result is a systematic release overshoot of approximately $0.5h$: released objects land, or are dropped from, a height proportional to half their own size, rather than fractions of a centimeter above the surface.

This bug was fixed once, in our VLM pipeline, by deriving `object_height` directly from measured depth ($h = \max(z_{\text{surf}} - z_{\text{top}}, h_{\min})$) rather than from a model-reported size. It then recurred independently in our LLM pipeline several weeks later, because that pipeline’s prompt still asked the model to compute `object_height` from its own scene-description arithmetic (`object_height = size[2]`), exactly the assumption the VLM-mode fix had already identified as wrong. Fixing the bug once, in one pipeline, did not fix it everywhere it existed. The eventual, general fix, described in §V, is to compute this quantity deterministically for *every* pipeline. The underlying error is not a mistake any particular prompt makes, after all, but a mismatch between what the model is asked to assume and what the physical gripper actually does.

System architecture diagram: task interface/GUI, the three planner nodes, the plan manager, the skill executor, and the code-grounding layer, with the debug-artifact and configuration paths annotated.

(placeholder; final figure to be supplied)

Fig. 1. RoboReason harness architecture: the Sense-Plan-Act pipeline of §III-B, with the code-grounding layer overriding model-computed geometry before execution.

TABLE I

GEOMETRY-GROUNDING FAILURE TAXONOMY. “SENSOR/ACTUATOR REQUIRED” MARKS THE PHYSICAL PROPERTY THAT MAKES EACH BUG IMPOSSIBLE TO OBSERVE IN A SYMBOLIC OR SCRIPTED SIMULATOR.

Bug	Symptom and root cause	Sensor/actuator required
Top-vs-mid-body grasp/release mismatch	Release-lift height set to an object’s full height, implicitly assuming a top-surface grasp; the executor actually descends mid-body ($d = \min(0.5h, d_{\max})$). Net effect: a systematic release overshoot of roughly half the object’s height. Recurred independently in two pipelines (VLM mode, then LLM mode).	Real gripper kinematics (mid-body descent is a deliberate stability choice, not a simulator default).
Double-counted stacking height	In a VLM→LLM hybrid pipeline, a depth-measured true top surface had the VLM’s separate, non-depth-verified size estimate added on top of it a second time by the downstream LLM’s own stacking formula ($\text{position.z} + \text{size}[2]$).	Real depth measurement of an already-correct surface height.
Footprint- vs. point-distance target matching	A release genuinely inside a tray’s real bounding box, but far from its registered center point, was treated as having missed the target zone by a point-distance check.	Real target geometry with non-trivial spatial extent (tens of centimeters), not a point.
Grasp-width sampled from the silhouette edge	Grasp width derived by deprojecting a bounding box’s left/right edge pixels, exactly on the object’s silhouette boundary, the single worst location for valid depth (occlusion/parallax). One failed edge sample aborted an otherwise valid plan.	Real depth noise at object boundaries; does not exist for a rendered or symbolic bounding box.
Width-dependent TCP offset	The gripper’s fingers pivot, so flange-to-contact distance is a function of how far apart they are commanded to open, not a constant, so it is wrong for any object whose width diverges from a single calibrated offset.	Real gripper mechanism (finger kinematics), absent from a symbolic action space.
Multi-object sequential-placement collision (open)	Zone-aware spacing keeps a target zone’s <i>final</i> release positions from overlapping, but does not prevent the arm clipping an already-placed object en route to the next release, nor a released object settling into a neighbor from residual momentum or a few cm of height error. Dominant failure mode in a 120-trial re-evaluation: every non-empty operator note (22/22) describes a collision or a fallen/displaced object during a multi-object task, in both the LLM and VLM pipelines.	Real multi-body contact between two already-placed objects and the arm’s approach trajectory, absent from any single-object placement check, symbolic or real.

B. Case 2: double-counted stacking height

Our static scene description uses the convention $\text{position.z} = \text{base reference}$, $\text{size}[2] = \text{height}$, so that $\text{position.z} + \text{size}[2]$ reconstructs a target’s top surface, the formula every LLM-mode prompt is given for computing a stacking release height. In our VLM→LLM hybrid pipeline, a VLM grounding call first deprojects a target’s pixel location to a real, depth-measured world coordinate, then hands the resulting scene off to the same LLM planner used in static-scene mode. An early version of this hybrid set the generated scene’s target.position.z directly to the depth-measured top surface, already correct and requiring no further adjustment, while leaving the VLM’s separate, unverified $\text{size}[2]$ guess untouched. The downstream LLM planner then applied its normal stacking formula, adding

the guessed height on top of an already-correct value and inflating every stack-on-target release by however wrong that guess happened to be.

The fix follows the same pattern as Case 1. Rather than trying to correct the VLM’s size guess, we derive the target’s real height from depth ($h = \max(z_{\text{top}} - z_{\text{surf}}, 0)$), with no minimum floor, since a flat target zone marked directly on the table should legitimately read as having near-zero height), then reset position.z back to a base reference so that the existing downstream formula reconstructs the true top surface instead of double-counting it. The quantity computed by sensor is inserted *into* the representation the model already expects, rather than trusted from the model’s own output.

C. Case 6 (open): multi-object sequential-placement collision

Unlike Cases 1–2, this bug is not closed by §V. When a task places several objects into the same target zone in sequence, stacking three cubes on a tray, say, or arranging four in a line on the table, each release’s *final* position is spaced to avoid overlapping the others already placed, by a grid layout clamped to the zone’s real bounds (§V). This closes the *static* version of the problem: no two final positions occupy the same footprint. It does not address the *dynamic* version. Nothing constrains the arm’s trajectory while approaching or releasing the n -th object to avoid clipping the $(n - 1)$ -th, already sitting in the zone, and nothing constrains a released object’s small residual motion (momentum, a few cm of height error) from displacing a neighbor after release. Both are genuinely three-dimensional, time-dependent contact problems, not resolvable by correcting a single scalar the way Cases 1–2 were.

We identified this bug from the same per-run debug artifacts as Cases 1–2. By the time it was isolated, though, it was already the dominant failure mode in the evaluation of §VI (22 of 22 non-empty operator notes, in both the LLM and VLM pipelines), rather than something we could re-run in isolation and close before reporting results. We report it here as an identified, partially-mitigated (by the static spacing fix) **open** problem, not a claimed contribution. §VII returns to what closing it would require.

D. The common pattern

The six bugs in Table I share a structure: a quantity that is trivially, exactly correct in a symbolic or scripted simulator becomes, on real hardware, (a) a sensor measurement subject to depth noise and occlusion, (b) a function of real actuator kinematics that a text-based action schema has no natural way to express, or (c), for Case 6, a consequence of real multi-body contact between two physical objects sharing space over time. None of these three failure modes can occur in an environment where object geometry is authored by the experiment designer or perfectly known by the physics engine. That, however, is precisely the setting in which LLM/VLM robot planners are most commonly benchmarked [5], [8].

V. CLOSING THE DISCOVERED FAILURES

A. Design principle

The fix for five of the six bugs in Table I follows the same rule: *pull geometric arithmetic out of the model and into code wherever the answer is computable from a sensor measurement or a paired action already in the plan, and leave the model responsible only for the symbolic decision of what to do and where, not how much.* The model still chooses which object to pick, which zone to release into, and in what order; a deterministic post-processing layer then overwrites the specific numeric fields most exposed to the failure modes of §IV, using only information already available at plan time (the paired pick action, measured depth, and the scene’s

registered target geometry). The sixth bug (Case 6, §IV-C) is not a scalar this rule can simply overwrite. It is a trajectory-level, time-dependent contact problem, and is addressed separately, and only partially, below.

B. Object and release height

Given a generated plan, `fix_object_height` recomputes every release step’s `object_height` as the true vertical distance from the paired pick’s actual grasp contact point to the table surface, $h - d$ in the notation of §IV-A, rather than accepting whatever height value the model reported. This directly closes Case 1: the release-lift amount is now derived from the *same* descent the executor actually commanded during the pick, which by construction cannot disagree with itself, instead of from a separate, independently-computed model output that could.

`fix_release_height` performs the analogous correction for the release *position*, and two design choices matter here. First, matching a release against the scene’s registered targets is done by **footprint containment** ($\text{position} \pm \text{size}/2$) rather than point-distance to a target’s registered center. Case 3 in Table I showed that point-distance matching misses valid placements inside a target zone that is tens of centimeters wide. Second, a release position that *exactly echoes* a target’s or object’s own registered coordinates is treated as intentional (a deliberate stack or placement) and left untouched, while a position that merely *drifted* into a zone’s footprint without echoing it is nudged back outside, along whichever axis requires the smaller correction. This happens only when that zone’s surface sits meaningfully above the bare table, though; nudging away from the near-table-height generic “table” zone would otherwise push objects off the table’s edge trying to escape a zone that legitimately contains most of the workspace.

C. Width-aware TCP offset

Because the gripper’s fingers pivot, the flange-to-contact distance for a given commanded width is not constant. We measure this relationship at three representative widths (fully closed, mid-open, fully open) and interpolate piecewise-linearly between them at plan time, setting the executor’s tool transform per-pick from the plan’s reported grasp width and resetting it to a default mid-open value after release. Pick descent is additionally capped at (calibrated offset – clamp clearance), so that tall objects are gripped nearer their top rather than driving the gripper’s rigid mounting body into them. This closes Case 5.

D. Zone-aware spacing: a partial mitigation for Case 6

For a target zone that receives multiple releases in one plan, we lay out occupants on a grid centered on the zone’s anchor point and clamped to its registered bounds, falling back to an unbounded grid only when no known zone matches. This closes the *static* half of Case 6: final release positions inside a zone no longer overlap, and no longer overflow a zone smaller than the model’s own (unclamped) spacing guess would have produced. It does not close the

dynamic half. The arm’s approach trajectory to the n -th release is not checked against the already-placed $(n - 1)$ -th object’s footprint, and a released object’s small residual motion is not constrained at all. §VI reports this as still the dominant failure mode; §VII returns to what closing the dynamic half would require.

E. Best-effort grasp-width sampling

For the VLM pipeline, grasp width is estimated by de-projecting a detected bounding box’s left and right edge pixels to measure real-world distance. Sampling directly on the object’s silhouette boundary is the single worst location for valid depth, since boundary pixels are the most exposed to occlusion and parallax error. A single failed sample previously aborted the entire batched deprojection call, crashing an otherwise valid plan over a parameter, grasp width, that does not even affect whether the *pick* and *release* coordinates themselves are safe. We fixed this by (i) sampling slightly inset from the raw silhouette edge rather than on it, landing on the object’s surface instead of its boundary, and (ii) isolating width estimation into its own best-effort call that degrades to the model’s own width guess on failure instead of aborting the plan. This closes Case 4 without requiring width to be depth-verified in every case, which is itself an open limitation we return to in §VII.

F. Validation

Each fix in this section was developed and iterated against real, previously-failing production runs: the per-run debug artifacts our system records for every planning call (raw model response, active configuration, and, in VLM modes, the captured camera frame) let us replay the exact inputs that produced a given hardware failure and confirm that the corrected pipeline now produces the hand-derived expected geometry for that same input, rather than validating only against synthetic test cases. This retrospective replay, together with a unit-level regression suite (45/45 passing across all reasoning-method implementations at the time of writing), gives us confidence that each fix addresses its diagnosed root cause. §VI reports the residual failure distribution with the harness’s code-grounding layer in place.

VI. REAL-HARDWARE EVALUATION

A. Design

With the deterministic grounding layer of §V in place, we ran a 120-trial real-hardware study comparing a static-scene LLM planner against a camera-grounded VLM planner, methodologically anchored to the task/environment complexity taxonomy and the TS/TSR/AETS metric family of Favali et al. [5], adapted to a physical rig. Six task conditions crossed three difficulty axes, task length, reference specificity, and affordance/perception load, each at an easy and a hard extreme, over a real scene of four color-distinguished cubes and a tray: *pp_easy/pp_hard* (pick-and-place, one vs. four objects, named directly), *sort_easy/sort_hard* (stacking and sorting), and *arith_easy/arith_hard* (the object to act on must be *computed* from a stated numeric

value mapping rather than named directly, the one condition designed to isolate reasoning load from perception load). Each condition was run 10 times per model, for 120 trials total, with a single reasoning method (chain-of-thought with self-consistency) held fixed across all trials, since it is reported as the most safety-robust strategy in [5] and avoids stacking reasoning-method variance on top of the LLM-vs-VLM comparison actually under test. Both the LLM and VLM planner use the same underlying model (`kimi-k2.6`) for text-only and image-grounded calls respectively. This is deliberate, so that grounding modality is the only variable isolated between arms, not a side effect of comparing two different models with different baseline capabilities.¹

B. Metrics, adapted for physical ground truth

Favali et al.’s TS (task safety), TSR (task success rate), and AETS (action efficiency on task success) metrics are fully automatic in a symbolic simulator, where whether an action “damages the environment” or a sub-task “completes” is always exactly derivable from simulator state. Our real UR5cb has no collision or force-fault sensor anywhere in its control path, and no vision-based verification of the final scene state. We therefore adapt the same three metrics to a **semi-automatic** protocol: everything the system can know automatically (the command, configuration, generated plan, and steps actually executed) is logged by the debug harness of §V for every trial; the operator supplies exactly two numbers immediately after each trial, was it safe, and how many sub-tasks completed correctly, from which TS, TSR, and AETS are computed identically to [5]’s definitions. This adaptation is itself a consequence of the paper’s central argument: full automation of these metrics is only possible because a symbolic simulator’s world state is always exactly known, which is precisely the property real hardware does not have.

At $n = 10$ per cell, point estimates alone invite exactly the kind of pushback a reviewer should raise, so every reported TS% and TSR% below carries a 95% Wilson score interval (TS is an exact per-trial proportion; TSR pools successes over $\text{sub_tasks_required} \times n$). One comparison was declared before any data was collected, not chosen afterward to fit the numbers: a two-sided Fisher’s exact test, LLM vs. VLM, on `arith_hard` full-trial success, at $\alpha = 0.05$. It is binarized per trial rather than pooled across a trial’s four sub-tasks, since those are not independent draws; one wrong decision tends to misplace every object in the trial together, not each independently. Everything else is reported descriptively only.

C. Results

Table II reports the headline result. Both models operate under the same deterministic grounding layer and the same underlying model, so the residual gap is neither a grasp-geometry gap (§V applies uniformly to both pipelines) nor a

¹An earlier run used different models per modality; the VLM-specific model was subsequently withdrawn from its provider’s catalog, forcing a full re-run on a single unified model instead.

TABLE II

OVERALL RESULTS, 60 TRIALS PER MODEL (120 TOTAL), WITH THE DETERMINISTIC GROUNDING LAYER OF §V IN PLACE.

Model	n	TS	TSR	AETS
LLM (kimi-k2.6)	60	98.3%	98.3%	0.1269
VLM (kimi-k2.6)	60	85.0%	89.2%	0.1149

Overall headline results — LLM vs VLM (n=60 per bar, 95% Wilson CI)

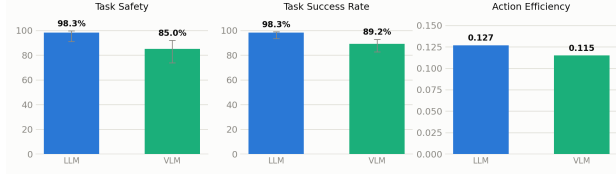


Fig. 2. Overall TS/TSR/AETS summary, LLM vs. VLM, with the deterministic grounding layer in place.

model-capability gap. It is specifically attributable to grounding modality. Breaking results down by task condition (Figure 3) shows the gap concentrated on the two hardest multi-object conditions, and predominantly in *safety*, not success rate: TS is 100% for both models on every easy condition and on pp_hard/sort_hard (LLM), but drops to 50% (VLM, sort_hard) and 60%/90% (VLM/LLM, arith_hard) on the two conditions with the most simultaneous in-zone objects. Neither model shows the sharp *success-rate* collapse a July pilot with different, modality-specific models had found on the arithmetic conditions specifically (30%/25% VLM TSR against 100%/97% LLM). With a single unified model, arith_easy is 100%/100% and arith_hard is 90%/85% (TSR, LLM/VLM): a real but modest gap, not a collapse. §VI-D returns to why. The one pre-registered comparison, Fisher’s exact test, LLM vs. VLM, arith_hard full-trial success, shows no detectable difference at this sample size (LLM 6/10, VLM 4/10, $p = 0.66$). That is consistent with a real but modest gap that $n = 10$ per cell cannot resolve to significance, not with an absence of one.

D. The failure mode is computation, not perception

Reading all 22 non-empty operator failure notes from the 120 trials, rather than trusting the aggregate numbers alone, was the key analysis step, as it was in an earlier pilot on this same task matrix. Every one of the 22 notes describes either a collision between objects (10) or an execution/grasp failure, a slipped grasp, a release too high, a dropped object (12), during a multi-object task (pp_hard, sort_hard, arith_hard). **Zero** notes describe a wrong object chosen for reasoning or value-mapping reasons, and zero describe an incomplete enumeration (Figure 4). This happens in **both** arms: the LLM contributes 4 of the 22 notes (all on arith_hard), the VLM the remaining 18, spread across all three multi-object conditions roughly in proportion to their trial counts. The failure recurs regardless of which planner grounded the scene, which is direct confirmation that Case 6 (§IV-C) is this taxonomy’s most consequential open item,

Safety and success rate by task — LLM vs VLM (n=10 per bar, 95% Wilson CI)

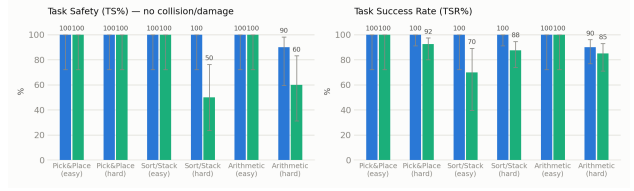


Fig. 3. Task safety and task success rate by task condition, with 95% Wilson CI.

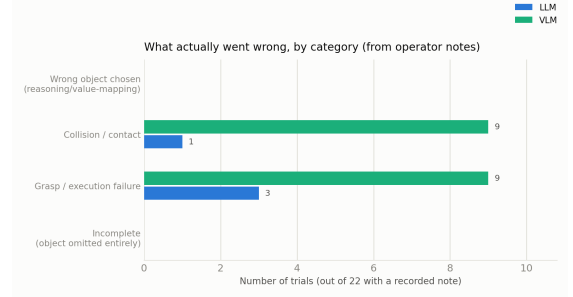


Fig. 4. Failure-mode breakdown, derived from the 22 non-empty operator notes out of 120 trials.

not a peculiarity of one pipeline.

Unlike the closed bugs of §IV, this result is not a second-order instance of the paper’s central claim; it is a direct one. Case 6 is itself one of the six geometry-grounding bugs of Table I; this evaluation is what identifies it as the dominant one once the other five are closed.

A note on an earlier finding. A pilot run on this same task matrix, using different, modality-specific models (not the single kimi-k2.6 used here), found a different dominant category: 13 of 31 notes read as a wrong-object choice concentrated specifically in the arithmetic condition, consistent with an off-by-one error in value-mapping reasoning under a stated numeric mapping. That finding does not replicate here: zero such notes this cycle, on every condition. We read the simplest explanation as the correct one: it was a property of the earlier VLM model specifically, not of VLM grounding in general, and does not survive a model change. We report it for the record and as a caution against generalizing a single-model finding to “VLM grounding” as a category, not as a result that still stands.

VII. DISCUSSION AND LIMITATIONS

Implication for evaluation practice. Symbolic and scripted simulation is necessary but not sufficient for validating an LLM/VLM robot planner: it validates reasoning strategy and task decomposition, but by construction cannot exercise the geometry-grounding failure class of §IV, since it never asks the model to compute a quantity a physical sensor could contradict. A geometry-grounding audit, identifying every numeric quantity a prompt asks the model to compute itself and asking whether a sensor or paired action already determines it more reliably, belongs alongside task-complexity benchmarking as a standard step before hardware

deployment.

The cost of the LLM/VLM-vs-VLA choice. §III-A argued for code-grounding over an end-to-end VLA on data-efficiency grounds. §IV is the other half of that trade: six bugs is the price of asking a model to compute physical quantities it was never trained to get right, and §V is what it costs to pay that price down. We read this as a cost worth paying, not a case against the architecture, since every bug in Table I was closeable once diagnosed, precisely because the pipeline stayed decoupled enough to diagnose it.

Case 6 remains the largest open item. Everything else in this paper is either closed or diagnosed precisely enough to know what closing it would take. Case 6 is neither: it is a trajectory-level, multi-body contact problem, and §V-D’s static-spacing mitigation does not touch its dynamic half. Closing it plausibly needs a motion-planning-level check of the approach trajectory against already-placed objects’ known footprints, or a settle-time verification step after each release so a displaced neighbor is caught before the next pick. We have not implemented or evaluated either, but the first is naturally an obstacle-avoiding approach skill on the harness of §III, treating already-placed objects the same way a mobile-base planner already treats static obstacles, and is the concrete next addition we have in mind.

Generalizability. Our evidence comes from one manipulator (UR5cb), one gripper, one camera rig, and a single underlying model (kimi-k2.6) used for both the LLM and VLM arm. We do not claim that the specific six bugs of Table I recur verbatim on different hardware, only that the *structural pattern*, geometry computed by a model instead of measured, and multiple real bodies sharing space over time, is a property of this architecture, not of this robot. Grasp width is the one quantity we still cannot cross-check against depth in every case, since a single top-down sample determines height unambiguously but not left-right extent; our best-effort edge-inset sampling (§V) reduces this gap without closing it.

Reliability of operator-supplied labels. Our semi-automatic protocol (§VI) rests on a single operator’s real-time judgment, unverified by a second, blinded annotator. This is not a hypothetical risk: an early pass over this cycle’s data marked every trial safe regardless of outcome, caught only by cross-checking the safety label against the operator’s own free-text note, and corrected before §VI’s numbers were computed.

VIII. CONCLUSION

We presented a real-hardware ROS2 harness for decoupled LLM/VLM manipulation planning, built on the premise that a model’s broad, non-robot-specific pretraining is usable for physical reasoning through code, without collecting task-specific action data, provided a deterministic layer grounds the plan’s numeric geometry in sensor measurement rather than trusting the model’s own arithmetic. Deploying that harness on a real UR5cb manipulator turned this premise into an observation: over six development sessions we found

six recurring geometry-grounding bugs, invisible to symbolic and scripted simulation by construction, and closed five of them with the harness’s code-grounding layer. A 120-trial real-hardware evaluation, using a single unified model for both grounding modalities, shows the sixth, a multi-object placement collision, as the dominant residual failure, occurring in both the LLM and VLM pipelines regardless of which one grounded the scene. Closing that bug’s dynamic, trajectory-level half as a new harness skill, along with a depth-verified treatment of grasp width, are the most direct next steps; more broadly, we argue that a geometry-grounding audit belongs alongside task-complexity benchmarking for any LLM/VLM harness headed to physical hardware.

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